Sentiment Analysis & Generative AI Pipeline

An end-to-end framework for data processing, Transformer fine-tuning, and LLM summarization.

TECHNICAL PROJECT REVIEW

01. Classification Model

Requirement : Transformer Fine-Tuning & Evaluation.

THE GRAY AREA PROBLEM

Ambiguous 3-Star Reviews

Standard mapping (1-2 neg, 3 neu, 4-5 pos) fails at the 3-star level. Sentiment in these cases is often mixed or context-dependent, leading to noise in the training data.

The Smart Filtering Approach

Isolated 3-star reviews and removed non-informative content (short/empty strings). This ensures the model only learns from high-quality linguistic candidates.

RoBERTa-Driven Inference

Used the 'twitter-roberta-base-sentiment-latest' model to compute Softmax probabilities. This identified reviews that are **genuinely** neutral based on language, not just stars.

Logical Grounding

The resulting hf_sentiment column serves as a ground truth that understands nuances, far exceeding the reliability of simple star ratings.

INFERENCE LOGIC

We leverage **Softmax Mathematics** to determine the most likely sentiment class.

By using a pre-trained models like **DistilBERT** and **RoBERTa**, we inherit a deep understanding of syntax and semantics, allowing us to filter candidates that represent the most valuable insights for our business model.

CLASSIFICATION STEP BY STEP

Data Preparation: Loading, cleaning, and combining review text.

Neutral Review Identification: Using a pre-trained Hugging Face model to **filter truly neutral reviews**.

Dataset Balancing: Creating a balanced dataset with an equal number of negative, neutral, and positive reviews.

Model Training: Fine-tuning RoBERTa-base for sentiment classification (0=negative, 1=neutral, 2=positive).

Evaluation and Comparison: Analyzing the performance of both models using key metrics and confusion matrices.

Asset Saving: Persisting the balanced datasets and trained models for future project stages.

FILTER TRULY NEUTRAL REVIEWS

Initial Filtering: We first filtered the entire dataset to keep only reviews with an overall rating of 3.0 (3 stars), as these are initial candidates for neutral sentiment. We also ensured the text was not empty or too short.

Sentiment Model Initialization: We used a pre-trained sentiment analysis model from Hugging Face, specifically cardiffnlp/twitter-roberta-base-sentiment-latest. This model is designed to classify text into 'negative', 'neutral', or 'positive' sentiments. Its tokenizer and the model itself were loaded.

Batch Prediction: A function predict_sentiment was defined to process the review texts in batches. This function tokenizes the text, feeds it to the loaded model, applies softmax to get probability scores for each sentiment, and then selects the label ('negative', 'neutral', or 'positive') with the highest probability.

FILTER TRULY NEUTRAL REVIEWS

Applying the Model: The predict_sentiment function was then applied to the 'text' column of the filtered 3-star reviews (df_neutral_candidates), and the predicted sentiment was stored in a new column called hf_sentiment.

inal Selection: Finally, we filtered these results again, keeping only the reviews where the hf_sentiment predicted by the model was 'neutral'. This subset of reviews formed our df_neutral_clean dataset, representing the 'true neutral' reviews.

DATASET BALANCE: THE KEY TO INSIGHT

57

9

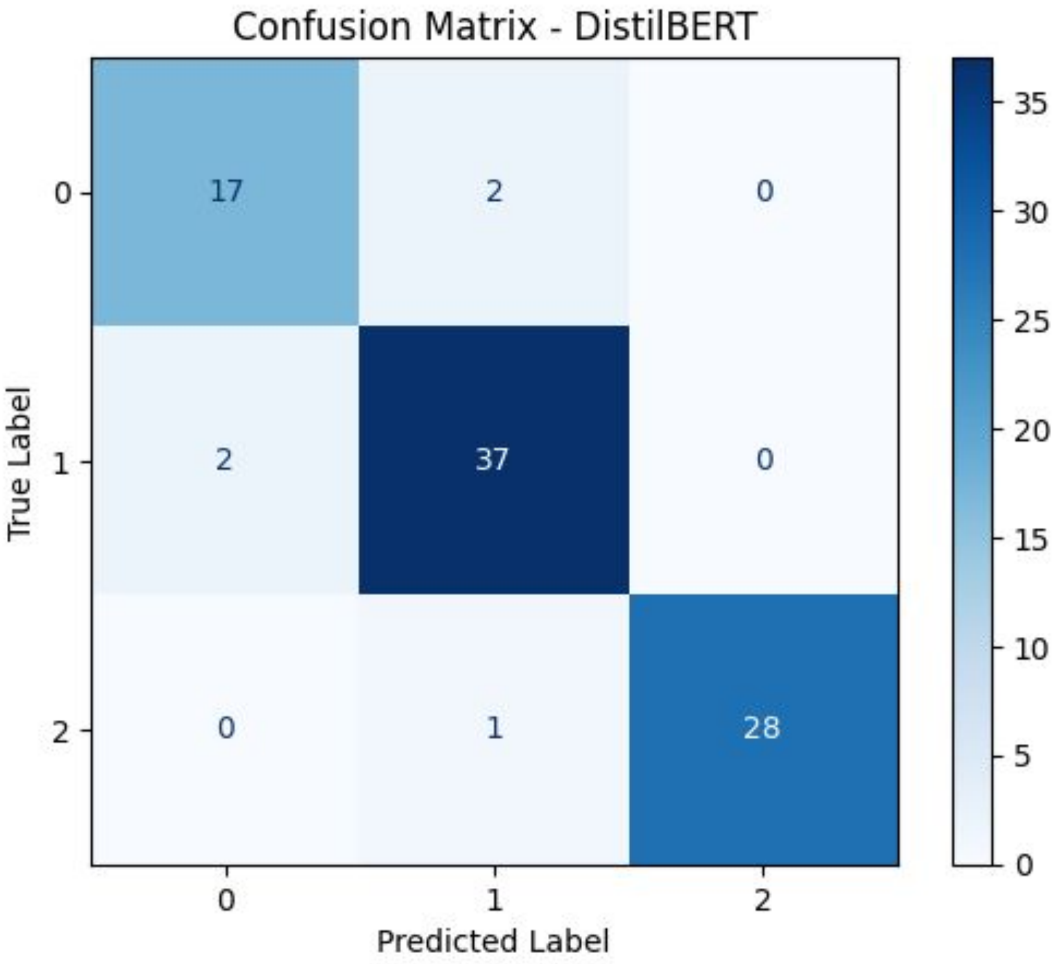
Total Examples

193	193	193
Positive	Neutral	Negative

Why this works?

Fine-tuning a pre-trained model doesn't require millions of samples. A perfectly balanced dataset prevents class dominance and forces the model to master the neutral class—the most valuable for real insights.

DATASET BALANCE 01: DistilBERT



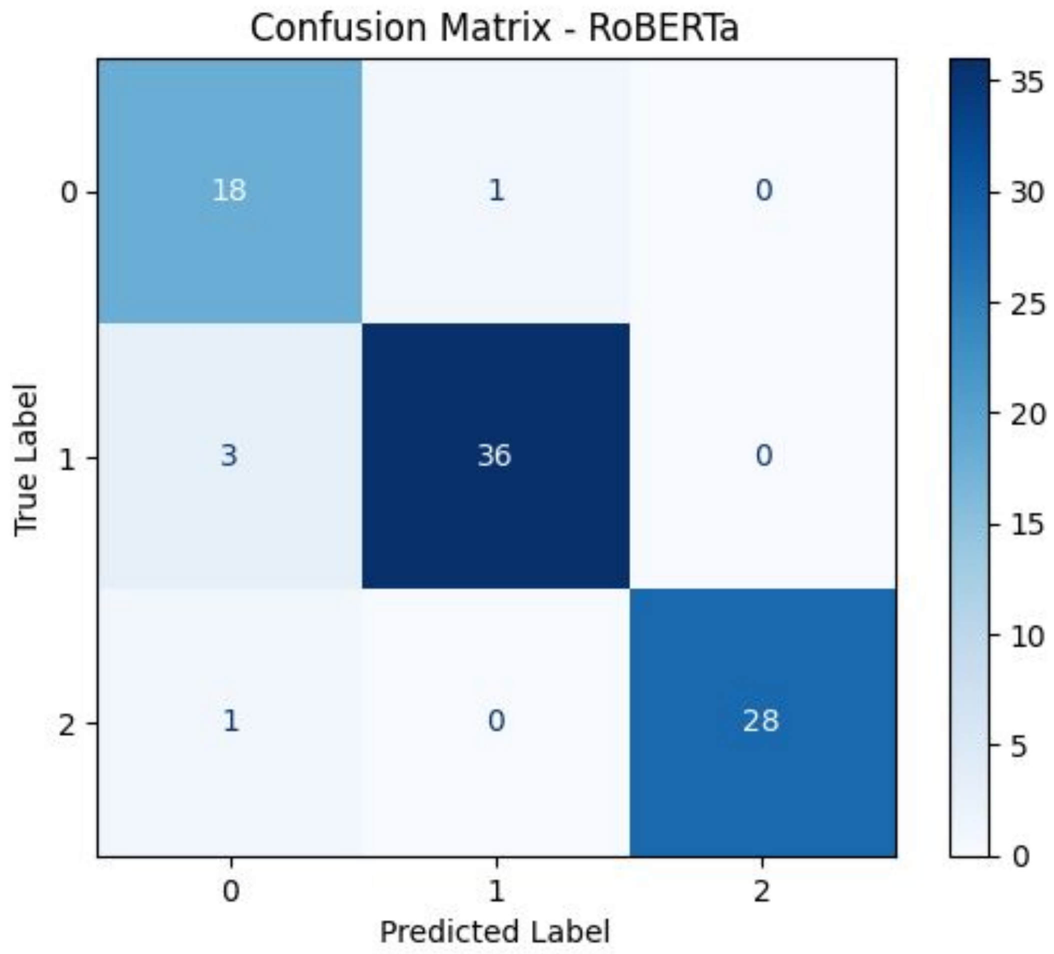
DistilBERT showed strong performance across all sentiment classes. The confusion matrix confirms high accuracy for negative, neutral, and positive reviews, with minimal misclassifications, mostly between similar negative and neutral texts. Positive reviews were rarely confused with negative ones. The model exhibits solid accuracy and balanced class performance.

```
Classification report:
              precision    recall  f1-score   support

           0     0.8947    0.8947    0.8947        19
           1     0.9250    0.9487    0.9367        39
           2     1.0000    0.9655    0.9825        29

    accuracy                         0.9425        87
   macro avg     0.9399    0.9363    0.9380        87
weighted avg     0.9434    0.9425    0.9428        87
```

DATASET BALANCE 01: RoBERTa



RoBERTa shows excellent performance, achieving 94% accuracy and a strong macro-F1 of 0.93. The confusion matrix confirms very few errors: negatives and positives are classified almost perfectly, and neutrals—typically the hardest class—are handled with high precision and recall. Overall, RoBERTa delivers a stable, reliable, and well-balanced sentiment classifier across all three classes.

```
Classification report:
              precision    recall  f1-score   support

           0     0.8182    0.9474    0.8780        19
           1     0.9730    0.9231    0.9474        39
           2     1.0000    0.9655    0.9825        29

    accuracy                         0.9425        87
   macro avg     0.9304    0.9453    0.9360        87
weighted avg     0.9482    0.9425    0.9439        87
```

02. Clustering & Summarization

Requirements: From Vectors to Narrative Insights.

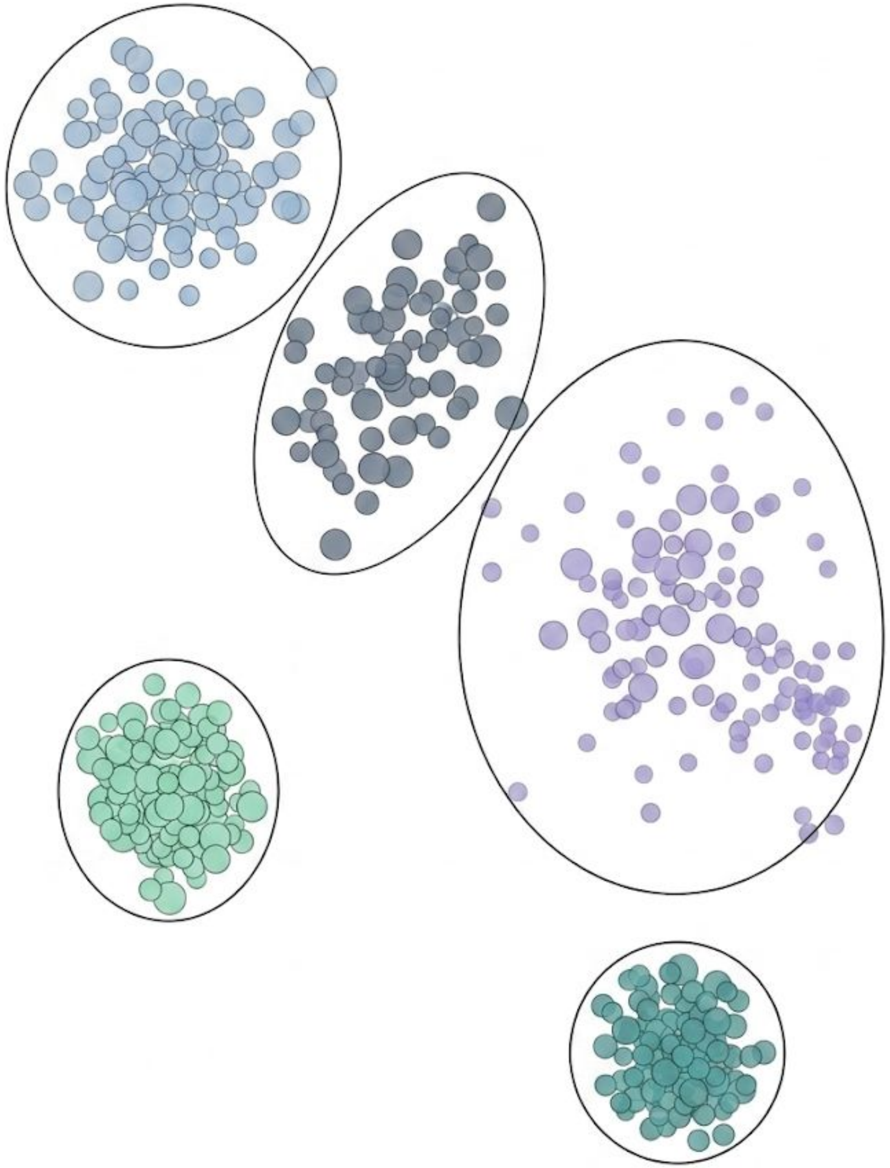
CLUSTERING

Data Preparation: Loading and balancing a dataset of Amazon reviews.

Embedding Generation: Using Sentence-BERT to convert review texts into numerical embeddings.

Dimensionality Reduction: Applying UMAP to reduce the complexity of these embeddings, making them easier to cluster and visualize.

Clustering: Employing both KMeans and HDBSCAN algorithms to group similar reviews into clusters based on their semantic content. The clusters are then visualized in 2D using **UMAP**.



WHY UMAP?

UMAP: Simplifying Complex Data - What is it?

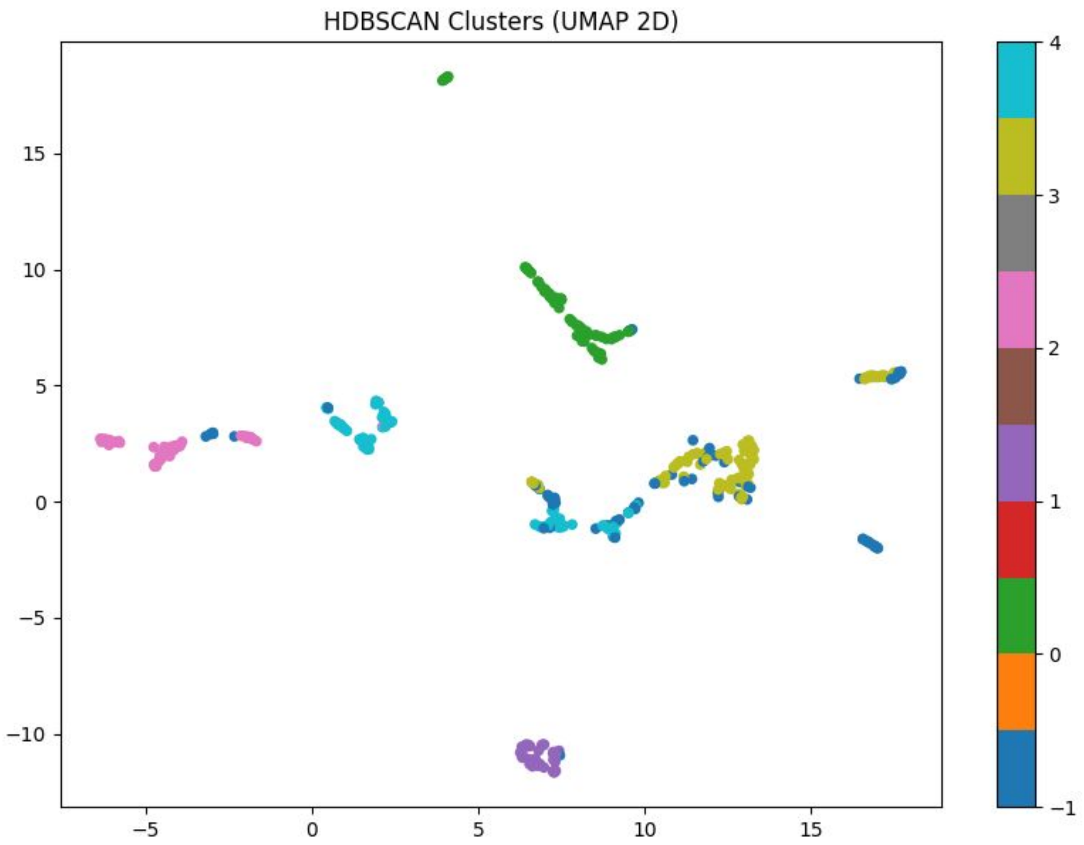
UMAP is a technique to simplify high-dimensional data (like our 384-dimension review embeddings) into fewer dimensions while preserving crucial relationships.

Why here?

Efficiency: Reduced 384D embeddings to 10D, making clustering faster and more effective.

Visualization: Further reduced to 2D to visually plot and understand the review clusters.

Achievement: Transformed complex review data into clear, actionable clusters and visual insights for better customer feedback analysis.



The UMAP plot shows review clusters in 2D. Points close together are reviews with similar meaning, while different colors represent different identified clusters.

CLUSTERING - 2.0

Cluster Analysis and Naming: Each cluster was analyzed by extracting example reviews, average ratings, and main products using Pandas and **value_counts()**. Descriptive names were assigned to the clusters for easier interpretation.

Export for Summarization: JSON files and text blocks were generated for each cluster, ready to be processed by Large Language Models (LLMs).

Detailed Analysis (Product/Category): Cluster information was integrated into the original dataset using Pandas.merge(), and a detailed analysis by product and category was performed, identifying top/worst products and their key metrics.

GENERATIVE AI SUMMARIZATION

This code generates structured prompts, enabling a Large Language Model (LLM) to perform detailed analyses of review data at different granularities. This approach ensures consistent and in-depth analytical outputs for reports and dashboards.

Each function takes a dictionary (representing a cluster, product, or category) as input and generates a comprehensive text prompt. These prompts are designed to instruct the LLM to act as an expert analyst and perform specific tasks.

GENERATIVE AI SUMMARIZATION

build_prompt_cluster

This function allows the LLM to analyze a review cluster, summarizing main themes, sentiments, key products, and actionable insights for that specific group.

build_prompt_product

This function enables the LLM to summarize a specific product, detailing its strengths, weaknesses, ideal user profile, and actionable recommendations based on its reviews.

uild_prompt_category

his function facilitates the LLM in analyzing an entire product category, identifying positive/negative themes, top products, and overall market positioning.

AI REVIEW INTELLIGENCE PLATFORM

This app analyzes Amazon product reviews to uncover meaningful insights. It classifies each review as positive, neutral, or negative, identifies the best and worst-performing product categories, and highlights the top products within each category based on sentiment trends.

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